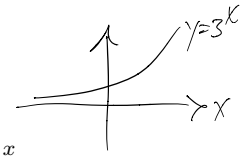
MATH 161: CHAPTER 6 PRACTICE

Name: Solutions

Complete the following problems using the ideas learned in Chapter 6 Inverse Functions, Exponentials, and Logarithms.

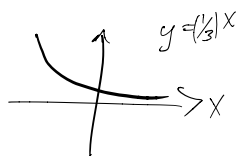
1. Find the limit: $\lim_{x\to\infty} 3^x$

∞



2. Find the limit: $\lim_{x\to-\infty} \left(\frac{1}{3}\right)^x$

∞



3. Find $\frac{dy}{dx}$. Assume y is a differentiable function of x.

$$3y = xe^{5y}$$

$$3y' = \frac{d}{dx}(x)e^{5y} + x\frac{d}{dx}e^{5y} = e^{5y} + x(5y')e^{5y}$$

$$\Rightarrow 3y' - 5xy'e^{5y} = y'(3 - 5xe^{5y}) = e^{5y}$$

$$\Rightarrow y' = \boxed{\frac{e^{5y}}{3 - 5xe^{5y}}}.$$

For problems 4-10, find $f'(x)$.

4. $f(x) = e^x \sin x$

$$f'(x) = \frac{d}{dx}(e^x)\sin(x) + e^x\frac{d}{dx}\sin(x) = e^x\sin(x) + e^x\cos(x) = \boxed{e^x(\sin(x) + \cos(x))}.$$

5. $f(x) = \ln(xe^{7x})$

$$f'(x) = \frac{\frac{d}{dx}\left(xe^{7x}\right)}{xe^{7x}} = \frac{e^{7x} + x\left(7e^{7x}\right)}{xe^{7x}} = \frac{e^{7x}(1+7x)}{xe^{7x}} = \boxed{\frac{1+7x}{x}}.$$

6. $f(x) = \dfrac{x}{\sqrt{1-(\ln x)^2}}$

$$\frac{d}{dx}\sqrt{1-\ln(x)^2} = \frac{1}{2}\left(1-\ln(x)^2\right)^{-\frac{1}{2}}\frac{d}{dx}\left(1-\ln(x)^2\right)$$

$$= \frac{1}{2\sqrt{1-\ln(x)^2}}\left(-2\ln(x)\left(\tfrac{1}{x}\right)\right)$$

$$= \frac{-\ln(x)}{x\sqrt{1-\ln(x)^2}}$$

$$\Rightarrow \quad f'(x) = \frac{\sqrt{1-\ln(x)^2} + x\left(x\dfrac{\ln(x)}{\sqrt{1-\ln(x)^2}}\right)}{1-\ln(x)^2}$$

$$= \frac{\dfrac{1-\ln(x)^2}{\sqrt{1-\ln(x)^2}} + \dfrac{\ln(x)}{\sqrt{1-\ln(x)^2}}}{1-\ln(x)^2}$$

$$= \boxed{\frac{1-\ln(x)^2+\ln(x)}{\left(1-\ln(x)^2\right)^{3/2}}}$$

7. $f(x) = (xe^x)^\pi = x^\pi(e^x)^\pi = x^\pi e^{\pi x}$

$$f'(x) = \boxed{\pi x^{\pi-1}e^{\pi x} + x^\pi\left(\pi e^{\pi x}\right)}$$

8. $f(x) = (e^{2x}+e)^{\frac{1}{2}}$

$$f'(x) = \frac{1}{2}\left(e^{2x}+e\right)^{-\frac{1}{2}}\left(2e^{2x}\right) = \boxed{\frac{e^{2x}}{\sqrt{e^{2x}+e}}}.$$

9. $f(x) = \left(\ln(5x^2+9)\right)^3$

$$f'(x) = 3\left(\ln(5x^2+9)\right)^2 \frac{d}{dx}\ln(5x^2+9) = 3\left(\ln(5x^2+9)\right)^2 \frac{\frac{d}{dx}(5x^2+9)}{5x^2+9}$$

$$= 3\left(\ln(5x^2+9)\right)^2\left(\frac{10x}{5x^2+9}\right)$$

$$= \boxed{\frac{30x\left(\ln(5x^2+9)\right)^2}{5x^2+9}}$$

10. $f(x) = \ln(5x^2+9)^3 = 3\ln(5x^2+9)$

$$f'(x) = 3\,\frac{\frac{d}{dx}(5x^2+9)}{5x^2+9} = \frac{3(10x)}{5x^2+9} = \boxed{\frac{30x}{5x^2+9}}$$

For problems 11-20, find the indefinite integral (you may need u-substitution).

11. $\int e^x dx = \boxed{e^x + C}$

12. $\int a^x dx$, where $a > 0$ is a constant ($\neq 1$).

$u = \ln(a)x$

$du = \ln(a)dx \Rightarrow \frac{1}{\ln(a)}du = dx$

$$\int a^x dx = \int \left(e^{\ln(a)}\right)^x dx = \int e^{\ln(a)x} dx$$

$$= \int e^u \frac{1}{\ln(a)} du = \frac{1}{\ln(a)}\int e^u du = \frac{1}{\ln(a)} e^u + C = \frac{1}{\ln(a)} e^{\ln(a)x} + C = \boxed{\frac{1}{\ln(a)} a^x + C}$$

13. $\int \pi^{2x} dx$

$u = 2x$

$du = 2dx$

$\Rightarrow \frac{1}{2}du = dx$

$$\int \pi^{2x} dx = \int \pi^u \left(\tfrac{1}{2}du\right) = \tfrac{1}{2}\int \pi^u du = \tfrac{1}{2}\left(\frac{1}{\ln(\pi)}\pi^u\right) + C = \boxed{\frac{1}{2\ln(\pi)}\pi^{2x} + C}$$

14. $\int \frac{1}{x} dx = \boxed{\ln|x| + C}$

15. $\int e^{2x} dx$ $\quad u = 2x,\ du = 2dx \Rightarrow \frac{1}{2}du = dx$

$= \int e^u \left(\frac{1}{2}du\right) = \frac{1}{2}\int e^u du = \frac{1}{2}e^u + C = \boxed{\frac{1}{2}e^{2x} + C.}$

16. $\int \frac{\ln x}{x} dx$ $\quad u = \ln(x)$, $du = \frac{1}{x}dx$

$\int \frac{\ln(x)}{x} dx = \int u\,du = \frac{1}{2}u^2 + C = \boxed{\frac{1}{2}\ln(x)^2 + C.}$

17. $\int \frac{\sqrt{\ln(x)}}{x} dx$ $\quad u = \ln(x)$, $du = \frac{1}{x}dx$

$\int \frac{\ln(x)^{1/2}}{x} dx = \int u^{1/2} du = \frac{2}{3}u^{3/2} + C = \boxed{\frac{2}{3}\ln(x)^{3/2} + C}$

18. $\int \frac{e^x}{\sqrt{1-e^x}}dx$

$u = 1-e^x$

$du = -e^x dx \Rightarrow -du = e^x dx$

$\int \frac{e^x}{\sqrt{1-e^x}}dx = \int \frac{-1}{\sqrt{u}}du = -\int u^{-1/2}du = -(2u^{1/2}) + C = \boxed{-2\sqrt{1-e^x} + C}$

19. $\int \frac{\ln(e^{2x})}{x^2}dx = \int \frac{2x}{x^2}dx$

$u = x^2$

$du = 2x\,dx$

$\int \frac{\ln(e^{2x})}{x^2}dx = \int \frac{du}{u} = \ln|u| + C = \ln|x^2| + C = \boxed{\ln(x^2) + C}$

20. $\int \frac{e^x}{3+e^x}dx$

$u = 3+e^x$

$du = e^x dx$

$\int \frac{e^x}{3+e^x}dx = \int \frac{1}{u}du = \ln|u| + C = \ln|3+e^x| + C = \boxed{\ln(3+e^x) + C}$

21. Evaluate the definite integral: $\int_2^3 \frac{xe^{x^2}}{3}dx$

$u = x^2$

$du = 2x\,dx \Rightarrow \frac{1}{2}du = x\,dx$

$\int_2^3 \frac{xe^{x^2}}{3}dx = \frac{1}{3}\int_4^9 e^u\left(\frac{1}{2}du\right) = \frac{1}{6}\int_4^9 e^u du = \boxed{\frac{1}{6}(e^9 - e^4)}$